

SEFIRE

THE SHARED EXPENSE TRACKER

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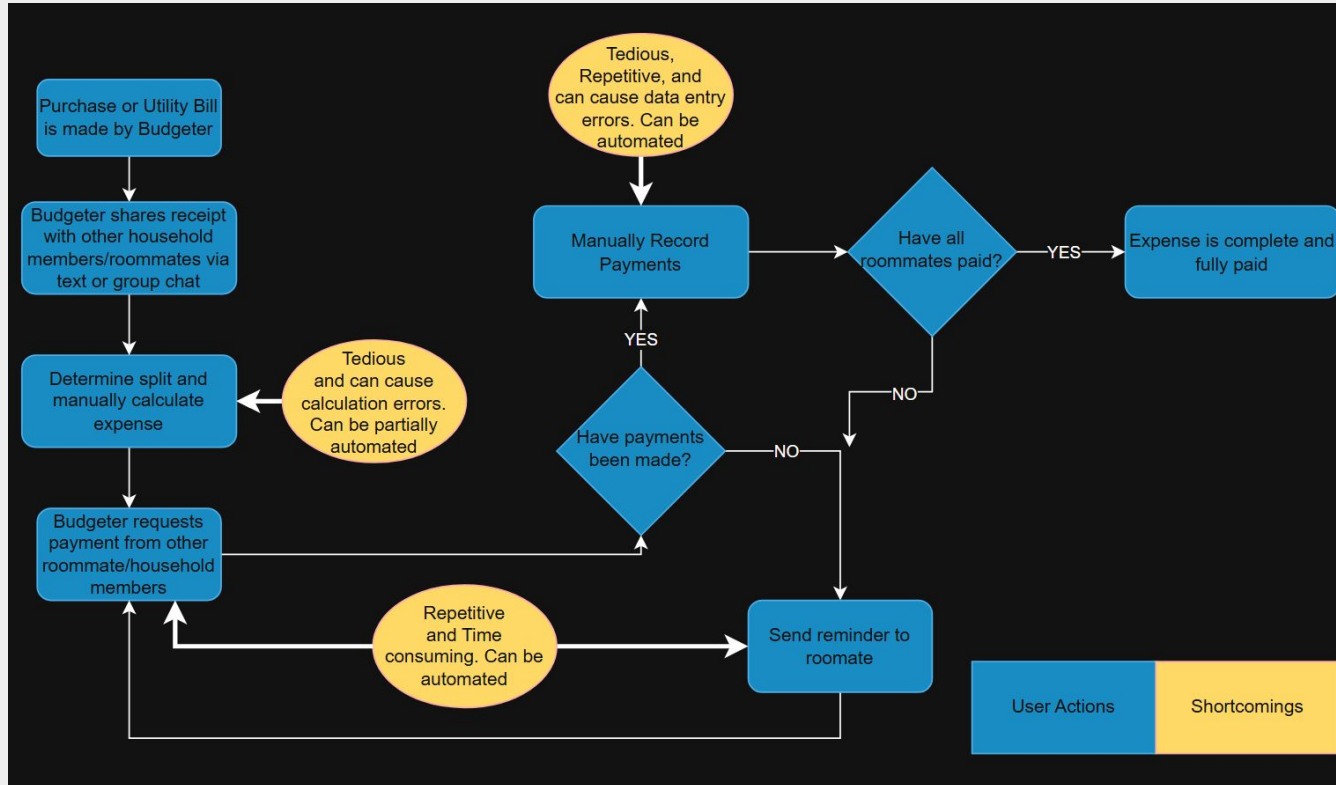


ELEVATOR PITCH

Rent and utility bills as well as groceries have to be shared by people living together in one home. However, it often gets very confusing trying to keep track of who paid what and who still owes money. The new app now keeps all shared expenses in one place. Users can then choose who is supposed to pay for what. They can always look at the updated balances and also note when payment has been made.

Feature that separates it from other applications: Be informative/resource on expenses and financial vocabulary

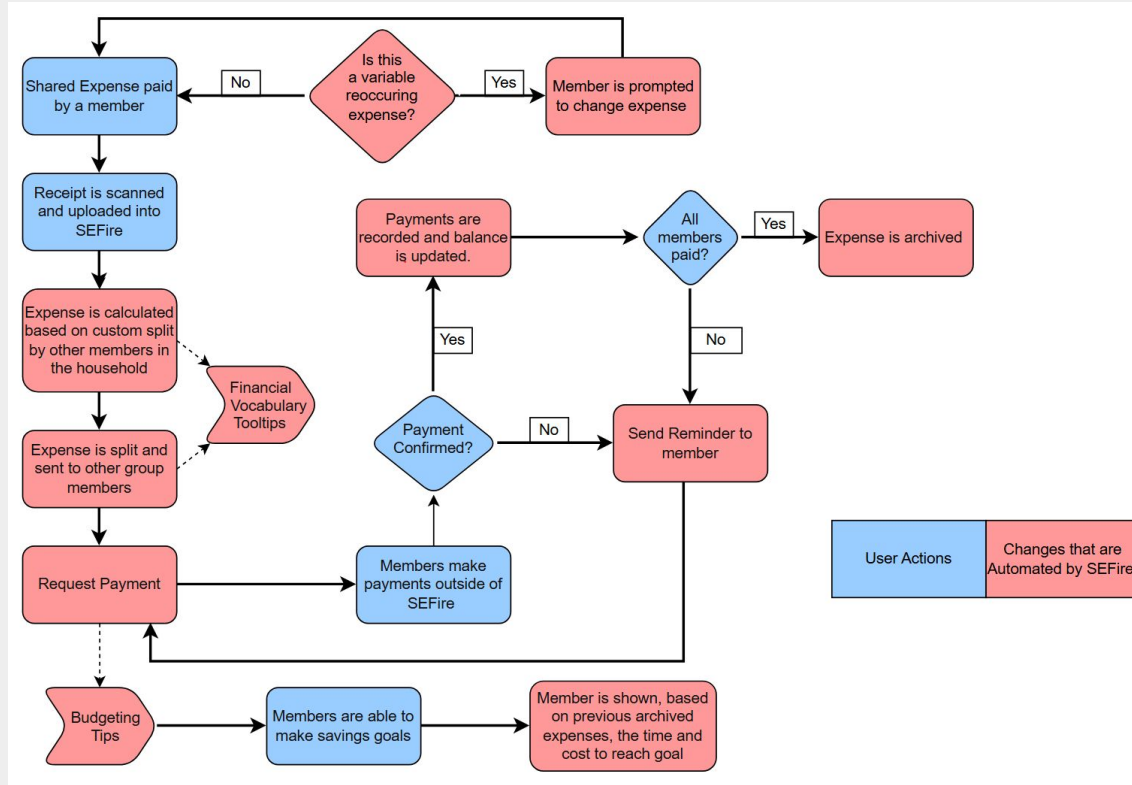
CURRENT (PROBLEM) PROCESS FLOW



SOLUTION STATEMENT

A mobile friendly app that help roomates record shared expenses, calculate each members shared expenses and track outstanding balances. The app also informs its members of the importance of budgeting and explaining financial vocabulary using helpful tooltips.

SOLUTION PROCESS FLOW DIAGRAM



SOLUTION PROCESS FLOW

Things that were added/changed

- **Variable Expense Prompt:** Members are prompted on whether the expense made has changed (if the expense reoccurs)
- **Custom Expense Splitting:** Expense is calculated using custom splitting made before the expense is added into the system.
- **External Payments:** Members must pay outside of the application as the application does not handle money within the app itself
- **Group Payment Verification:** Since the app considers multiple members in a group, it verifies all members have paid their expense (if necessary) before archiving the expense as completely paid.
- **Budgeting Awareness:** Members are informed of the importance in having a budget and provide tips to help manage their expenses.
 - Members are also able to place a savings goal to work towards
- **Financial Vocabulary:** Members can view definitions of unfamiliar financial terms via tooltips.

WHAT IT WILL DO

- Create a household group and invite members.
- Record bills with payer, total, and member sharing cost.
- Custom Splitting Expenses
- Custom Variable Expenses (Expected vs Actual)
 - Allows the input of “variable income” to determine expense expectation by next paycheck?
- Itemizes receipts
- Show balances
- Records payments
- Send reminders for upcoming bills/subscriptions
- Indicator/reminder for individuals who has not paid for a specific expense

WHAT IT WILL NOT DO

It won't pay your bills for you. (Won't automatically receive bills)

Force someone to pay.

No in app payments/No payment method connected.

- Banking information will not be stored
- Will not facilitate transactions between users

Manage nor teach people their Personal Finances

Manage Investments

Cook for you (this is household expenses not household chores)

COMPETITION MATRIX PART 1

Functionality	SEFire	Splitwise	Tricount	Settle Up	SplitCircle
Invite members / Create Groups	✓	✓	✓	✓	✓
Record shared bills	✓	✓	✓	✓	✓
Savings Goals via Budget tooltips	✓	✗	✗	✗	✗
Custom splitting expenses	✓	✓	✓	✓	✓
Variable Expense Prompt	✓	✗	✗	✗	✗
Show balances	✓	✓	✓	✓	✓

COMPETITION MATRIX PART 2

Functionality	SEFire	Splitwise	Tricount	Settle Up	SplitCircle
Record payments	✓	✓	✓	✓	✓
Send reminders	✓	✓	✓	Premium	✓
Notifications for those who have not paid	✓	✓	✓	Premium	✓
Financial Vocab/Tooltips	✓	✗	✗	✗	✗
Scan Receipts (OCR)	✓	Premium	✗	✗	Premium

DEVELOPMENT TOOLS

- **Integrated Development Environment (IDE)** - VSCode
- **Version Control** - Git through GitHub.
- **Continuous Integration (CI) and Continuous Deployment (CD)** - GitHub Actions & Workflows
- **Frontend/UI** - React Native
- **Backend** - Supabase
- **Database** - PostgreSQL
- **Selected Language (Backend)** - SQL (PostgreSQL and Supabase), Python (Supabase), and JavaScript (for React Native)
- **Selected Language (Frontend)** - A combination of HTML, CSS, and JavaScript (webpage)
- **Testing Framework(s)** - PyTest for Python and Jest for JavaScript
- **Documentation Tool** - dbForge for PostgreSQL, pydoc for Python, and JSDoc for JavaScript

MAJOR FUNCTIONAL COMPONENTS

User Interface: Mobile phone application and web browser, users can enter expenses, check balances, and track savings

Web/App Interface: React Native, renders SEFire's interface across mobile app and web instead of separate applications

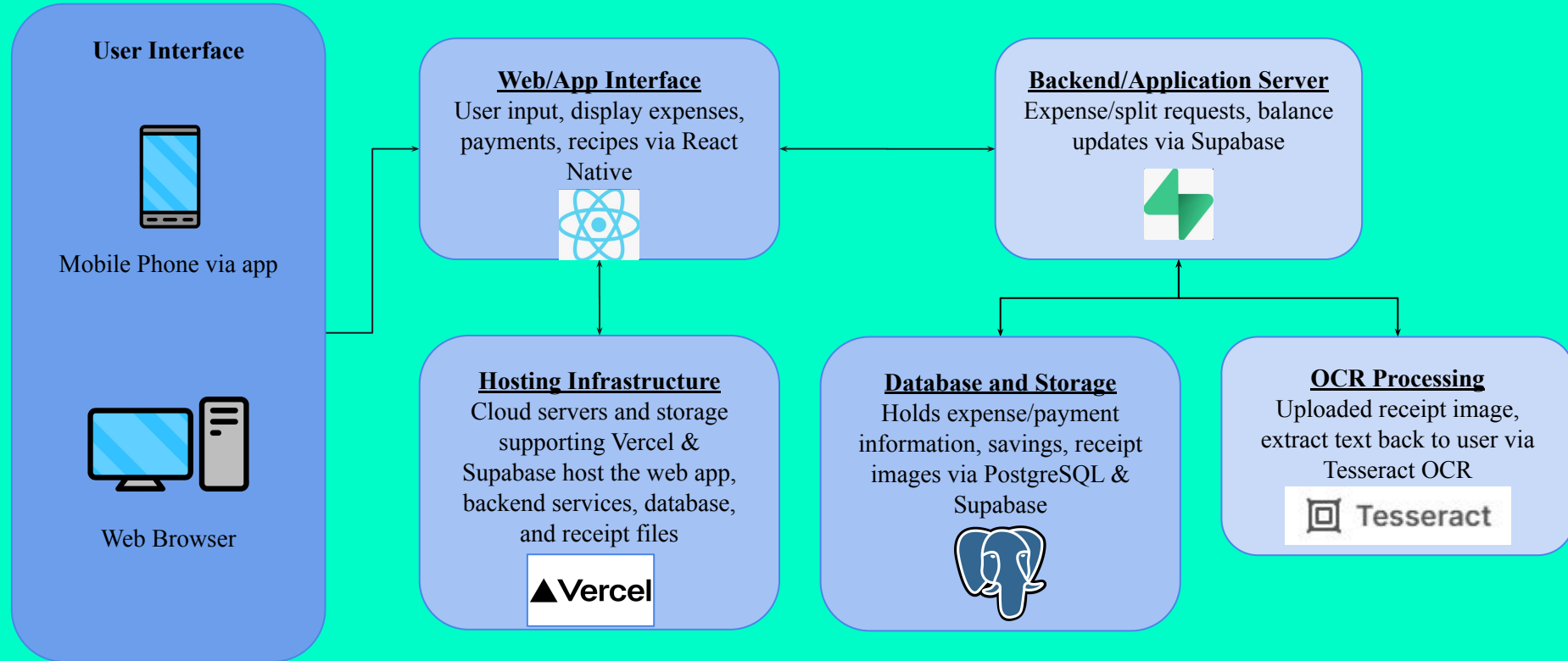
Backend: Supabase handles user accounts, application logic for expenses, splitting costs, and updating balances

Database: PostgreSQL stores household membership, expenses, payment records

OCR Processing: Tesseract OCR extracts details from uploaded receipts, extracted information can be edited and saved

Hosting Infrastructure: Cloud servers and storage supporting Vercel and Supabase. Hosts the web application, backend services, database, and receipt files

MAJOR FUNCTIONAL COMPONENTS DIAGRAM



RISKS

- 1.) **Customer and End User:** Low User Adoption
 - a.) Users may prefer existing methods or competing expense-management apps.
- 2.) **Technical:** OCR Inaccuracy/ Integration Issues
 - a.) Receipt data may be extracted incorrectly, and software components may encounter integration issues.
- 3.) **Security:** Exposure of User Financial Data
 - a.) Unauthorized access could expose users' financial or household information.
- 4.) **Legal:** Privacy and Data Handling Requirements
 - a.) User information and uploaded receipts must be stored and handled appropriately.
- 5.) **Financial:** Hosting & Service Costs
 - a.) Growth in users could increase database, hosting, and third-party service costs.

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APPENDIX

REAL WORLD VS PROTOTYPE TABLE

Features & Functionality	RWP	Prototype (Planned)	Prototype (Actual)